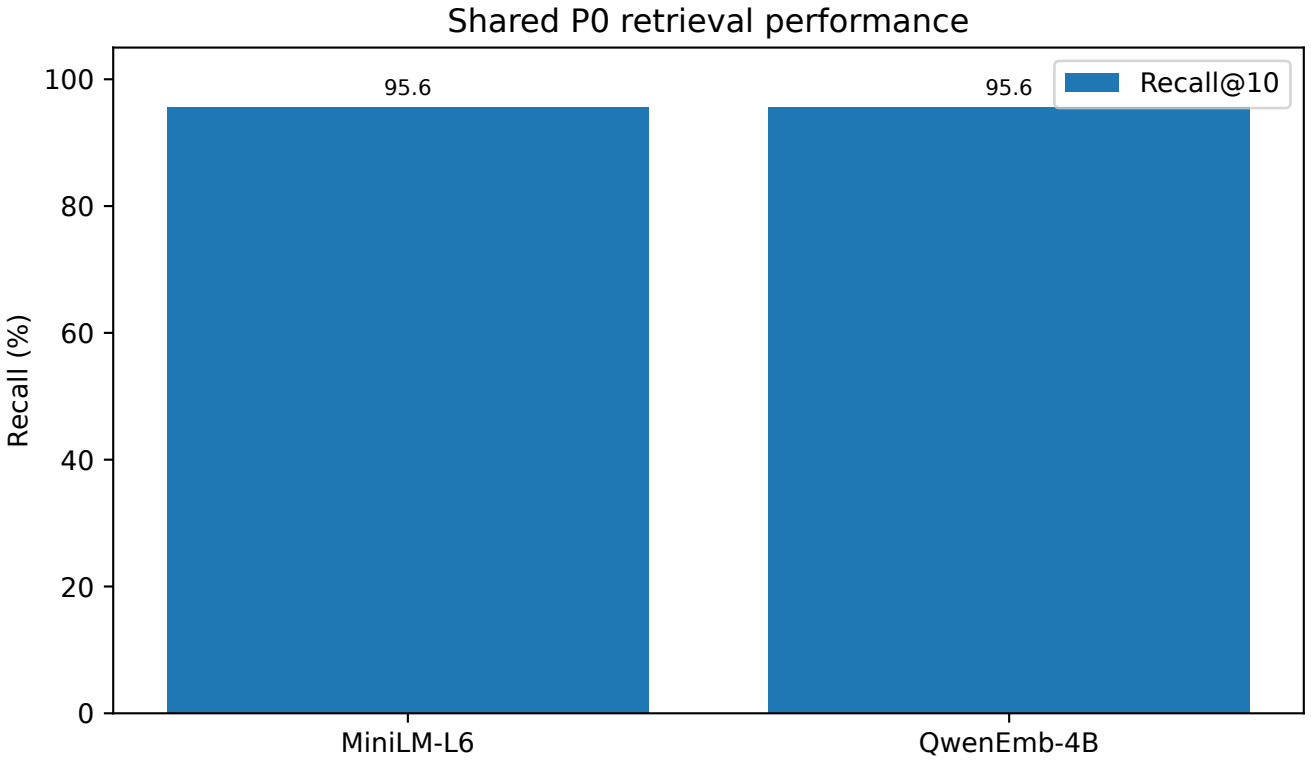
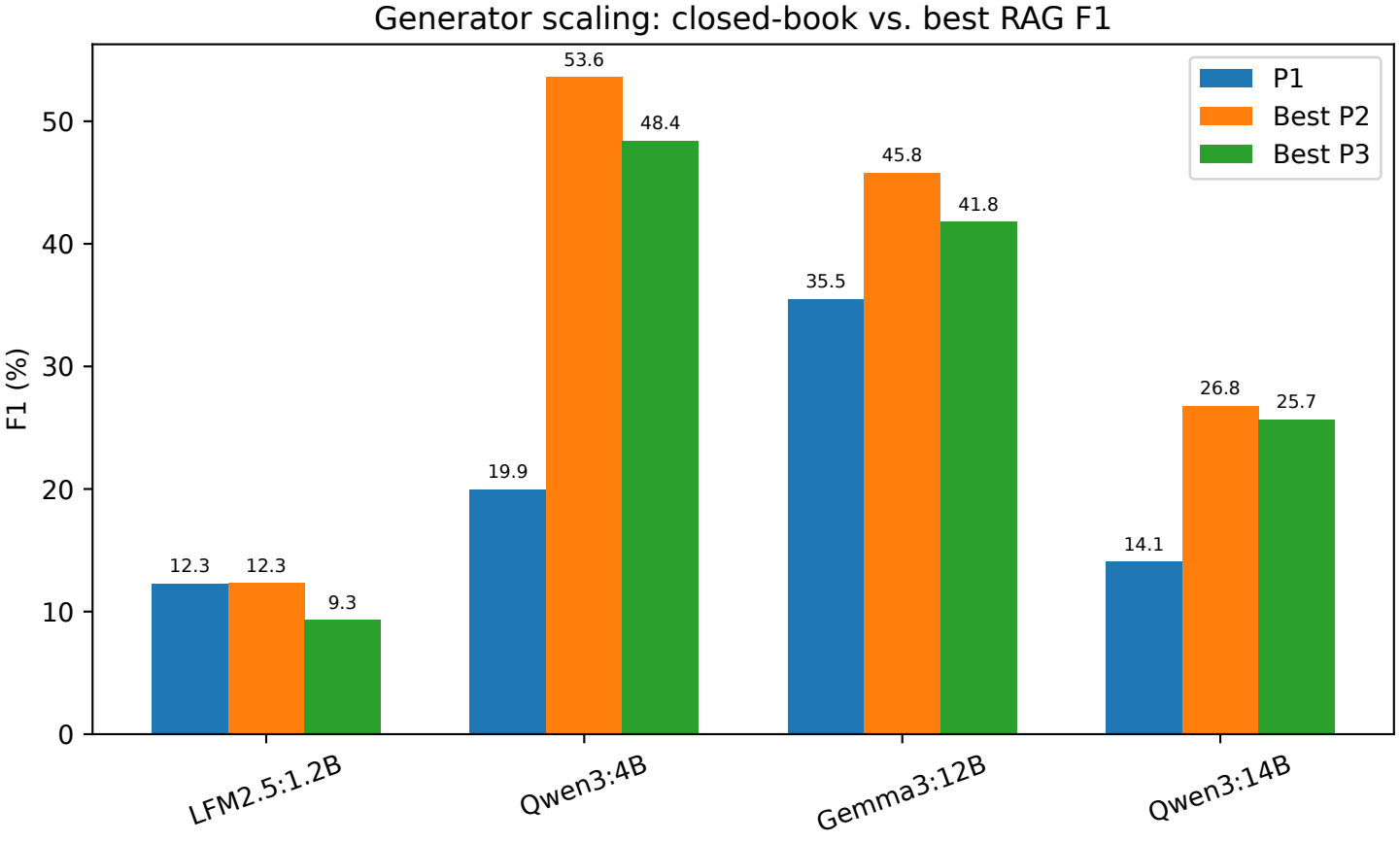
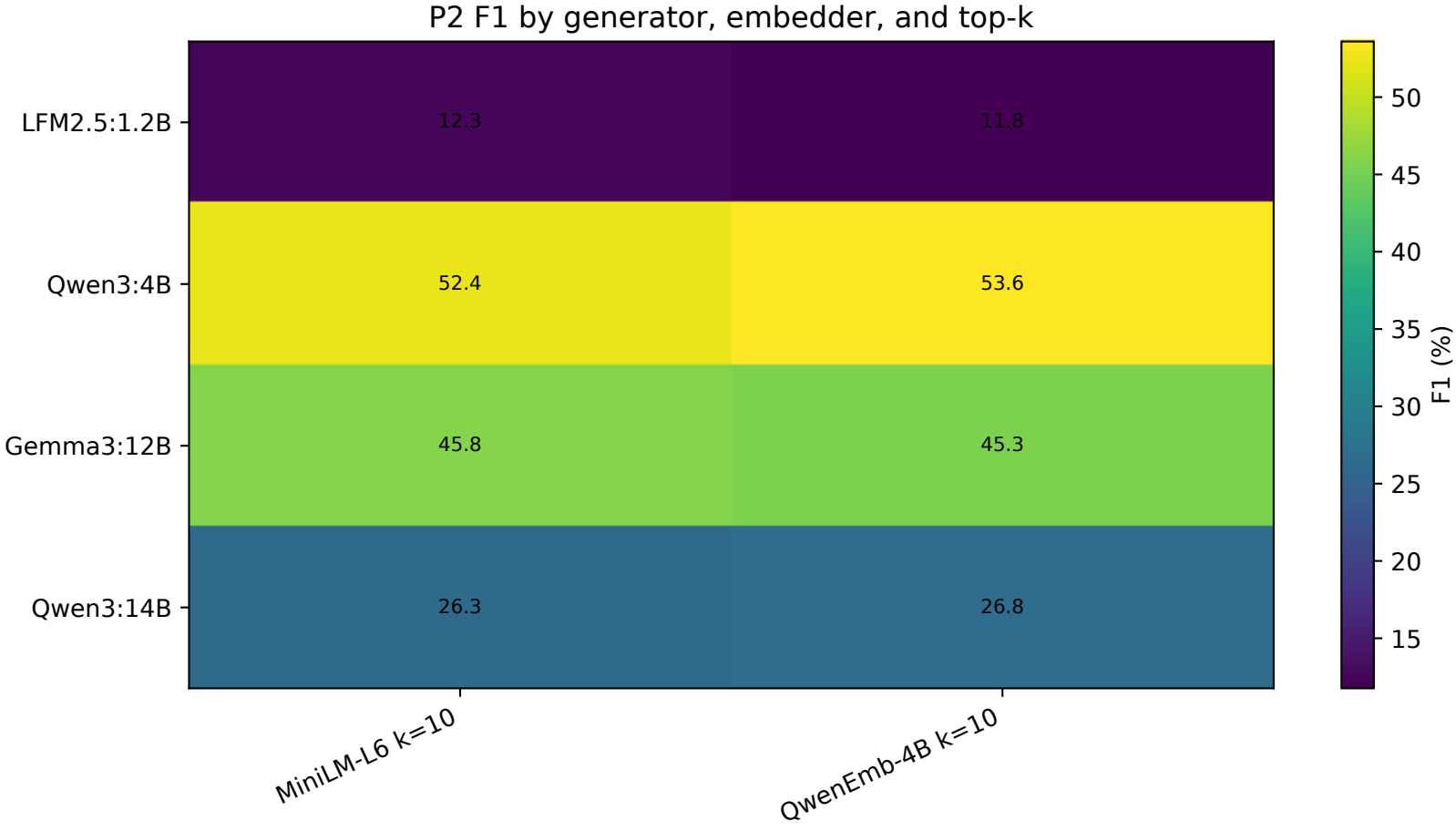
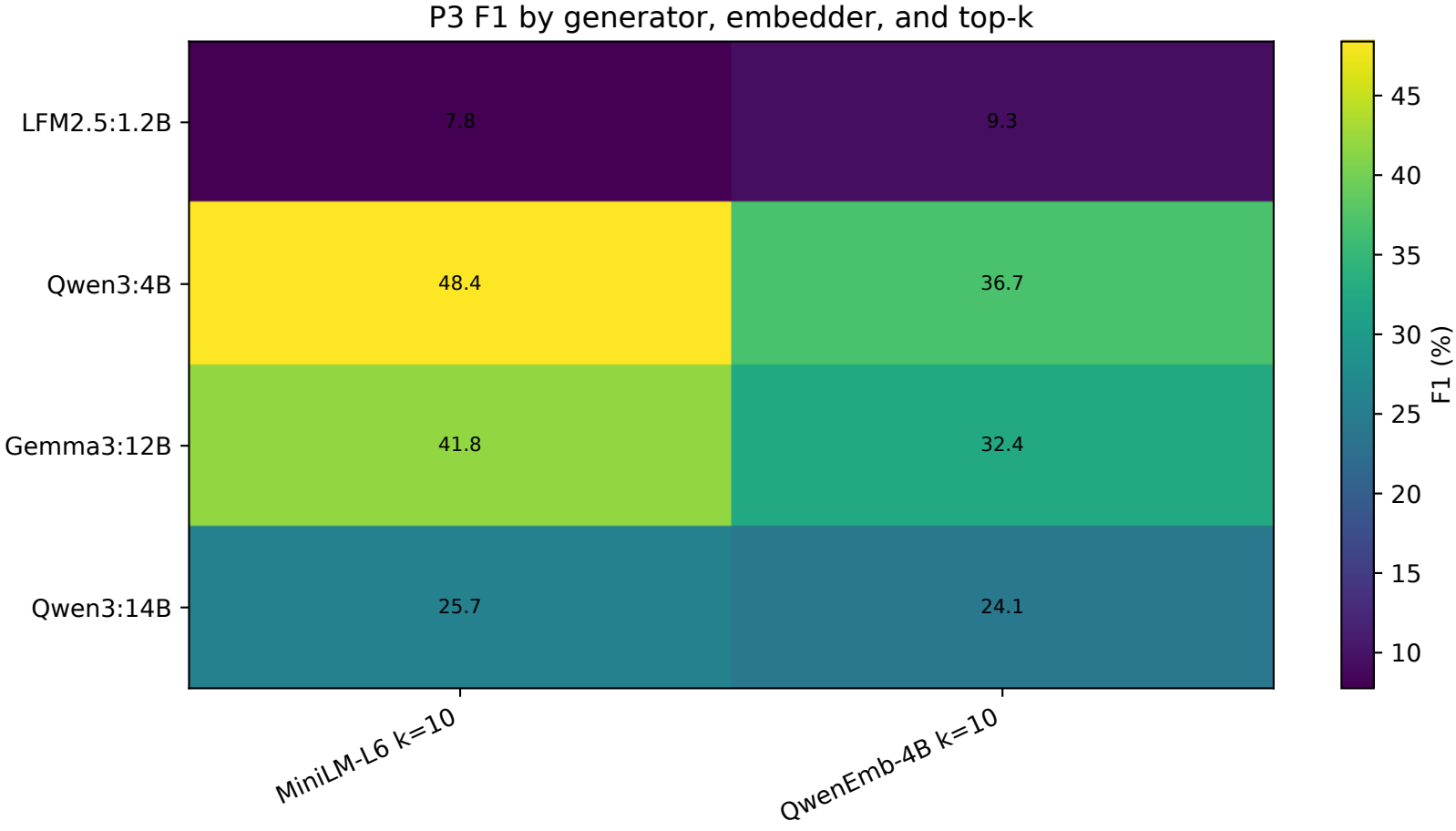
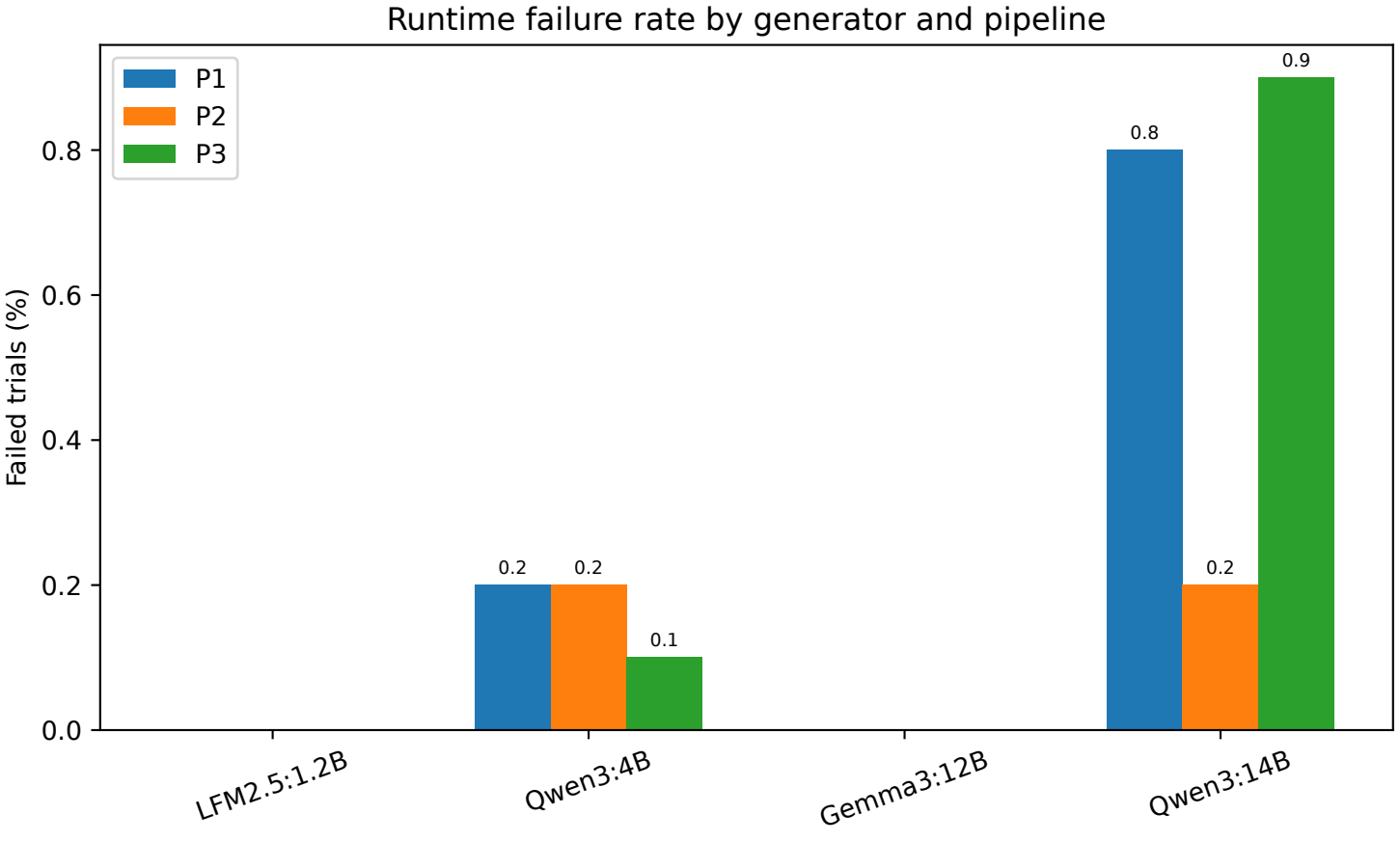
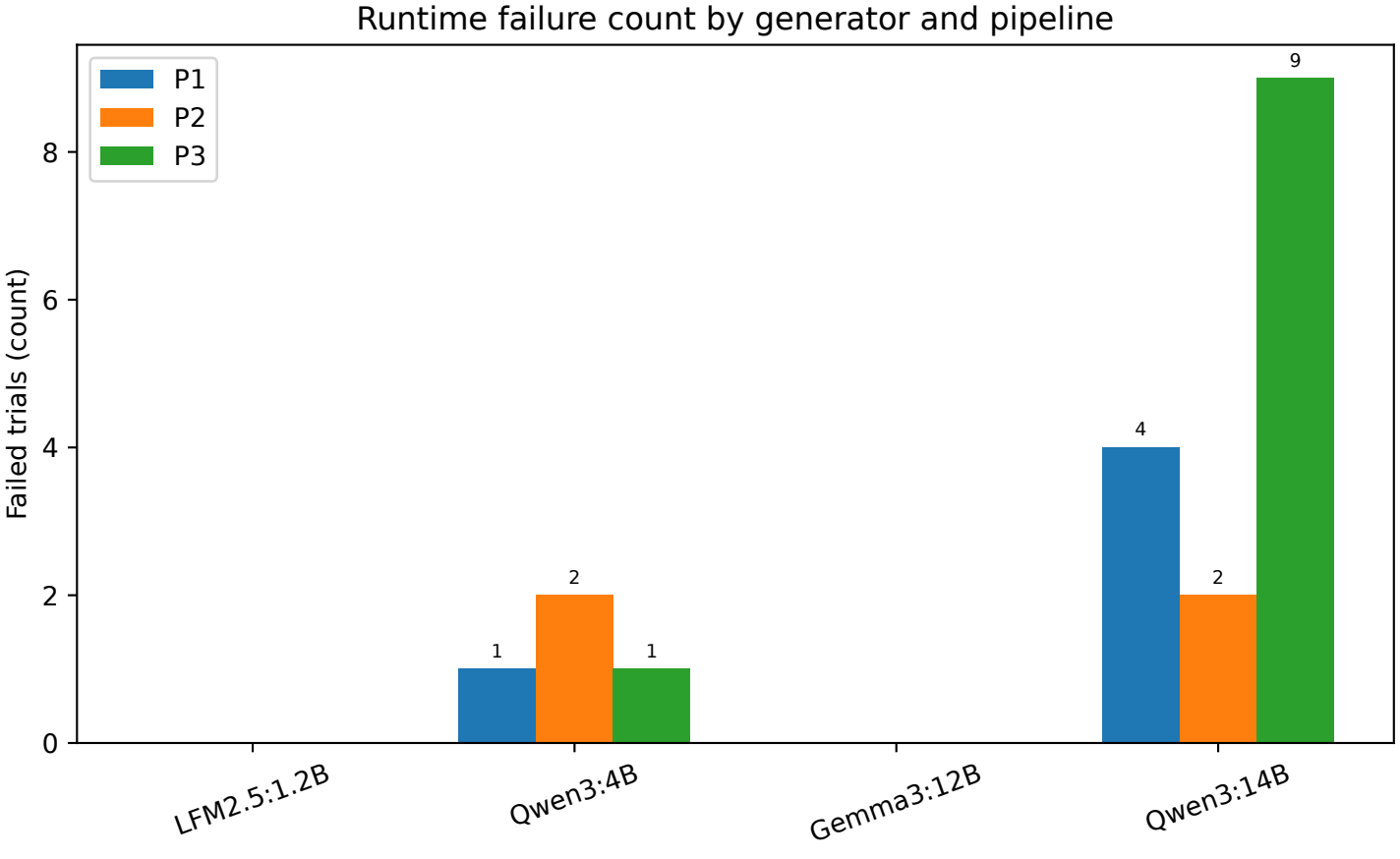

Shared P0 retrieval performance
Recall@10
95.6
95.6
Recall (%)
100
80
60
40
20
0
MiniLM-L6
QwenEmb-4B

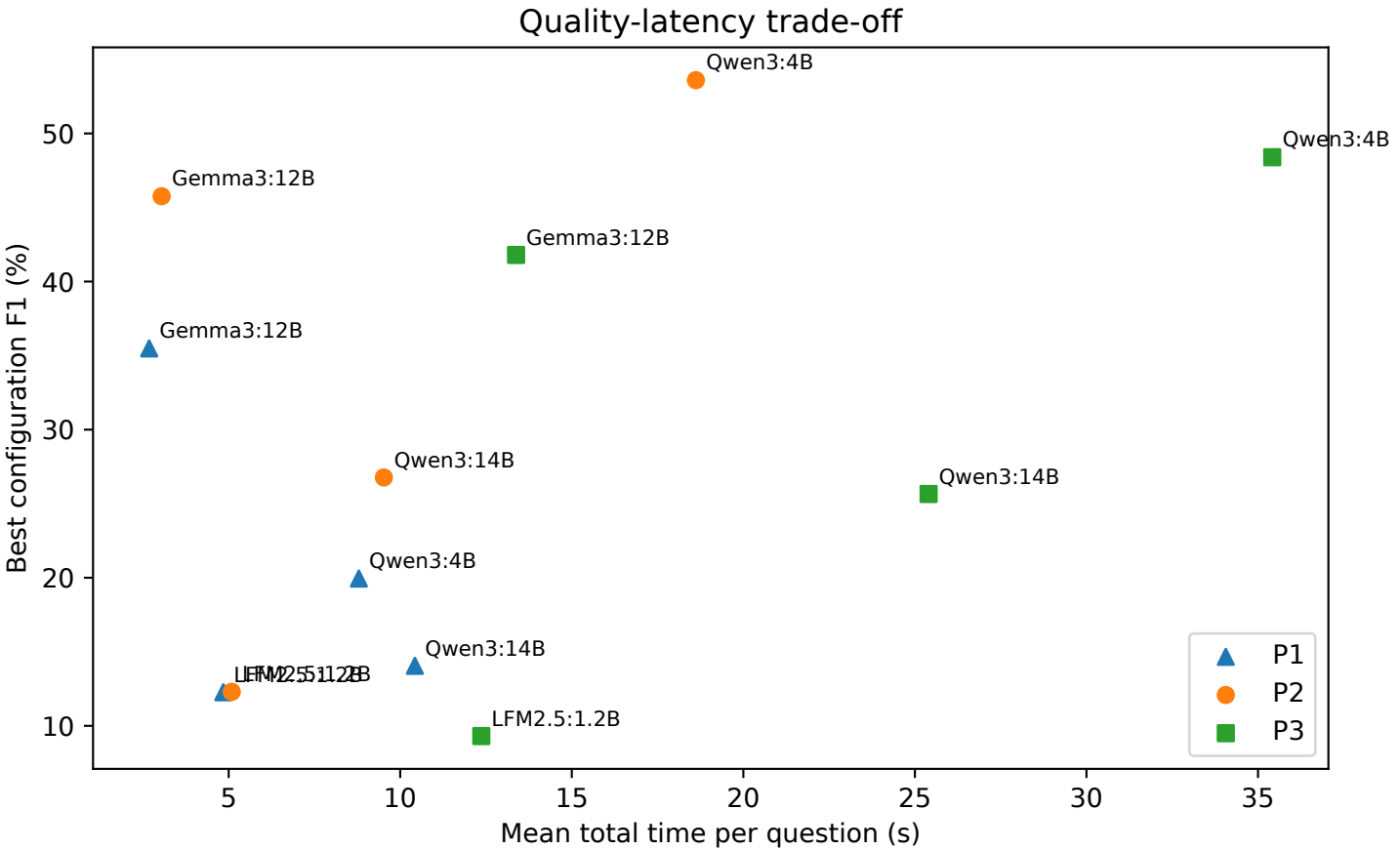

Generator scaling: closed-book vs. best RAG F1
F1 (%)
0
10
20
30
40
50
P1
Best P2
Best P3
12.3
12.3
9.3
19.9
53.6
48.4
35.5
45.8
41.8
14.1
26.8
25.7
LFM2.5:1.2B
Qwen3:4B
Gemma3:12B
Qwen3:14B

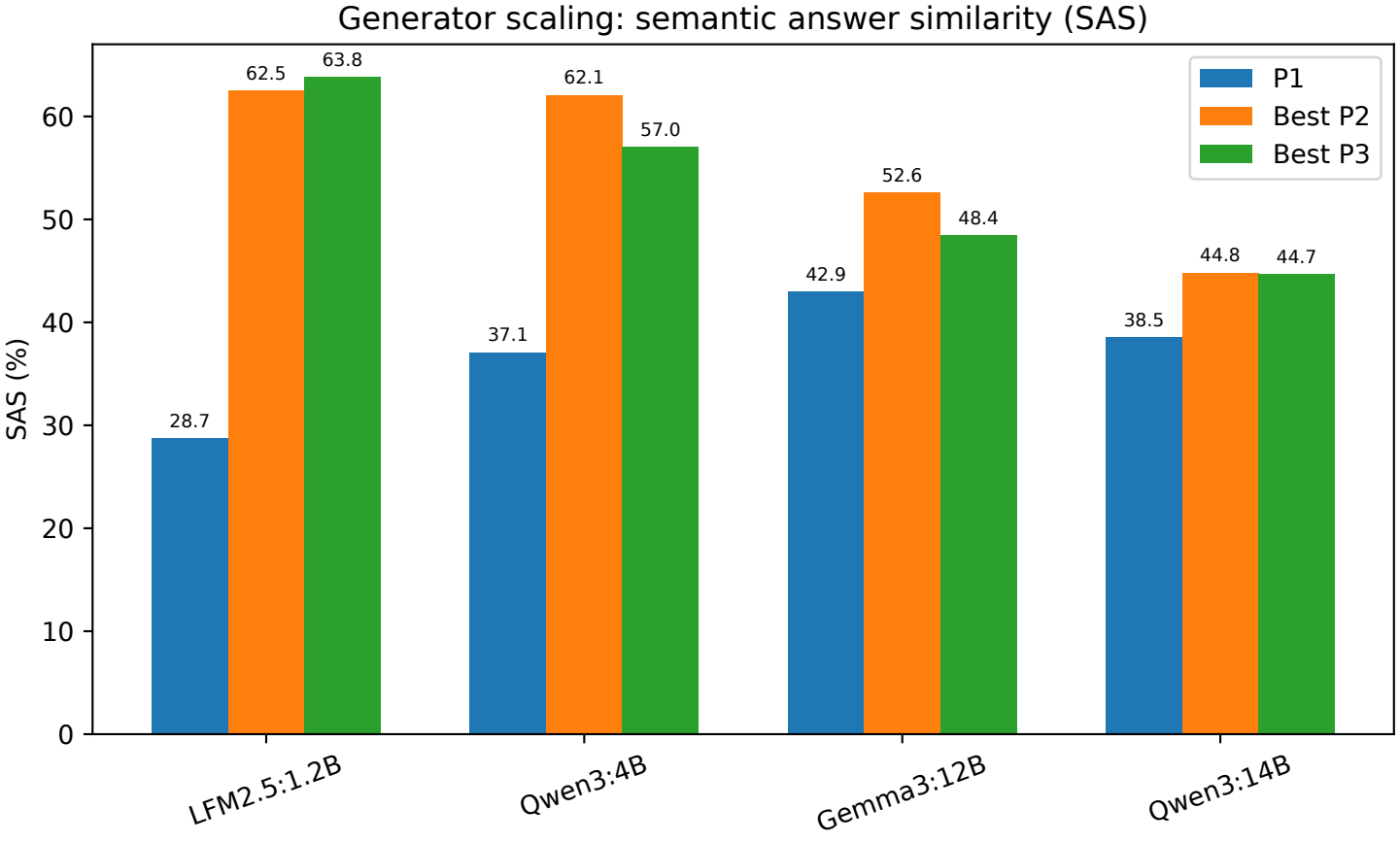

P2 F1 by generator, embedder, and top-k
LFM2.5:1.2B
Qwen3:4B
Gemma3:12B
Qwen3:14B
12.3
11.8
52.4
53.6
45.8
45.3
26.3
26.8
MiniLM-L6 k=10
QwenEmb-4B k=10
F1 (%)
50
45
40
35
30
25
20
15

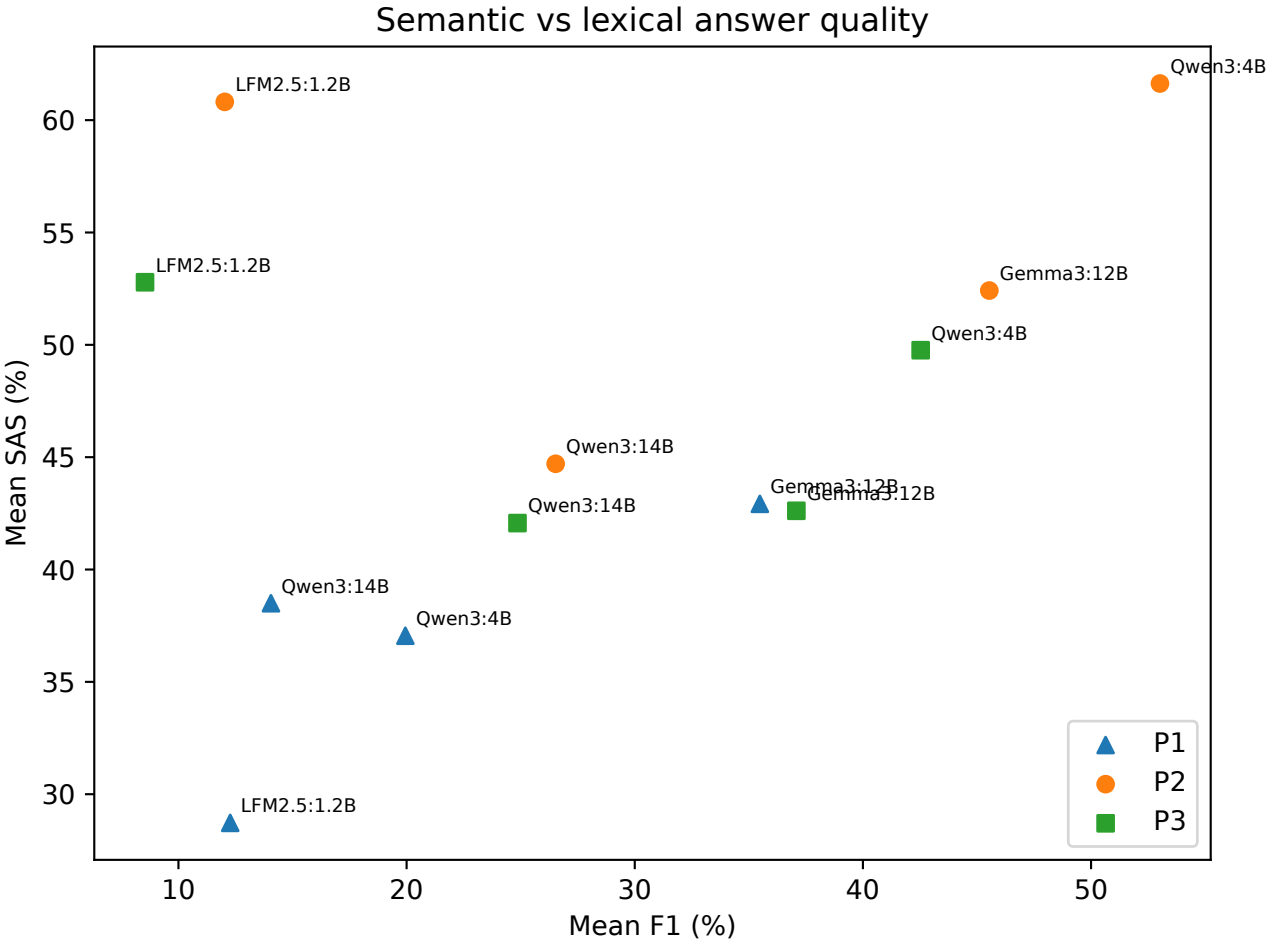

P3 F1 by generator, embedder, and top-k
LFM2.5:1.2B
Qwen3:4B
Gemma3:12B
Qwen3:14B
7.8
9.3
48.4
36.7
41.8
32.4
25.7
24.1
MiniLM-L6 k=10
QwenEmb-4B k=10
F1 (%)
45
40
35
30
25
20
15
10

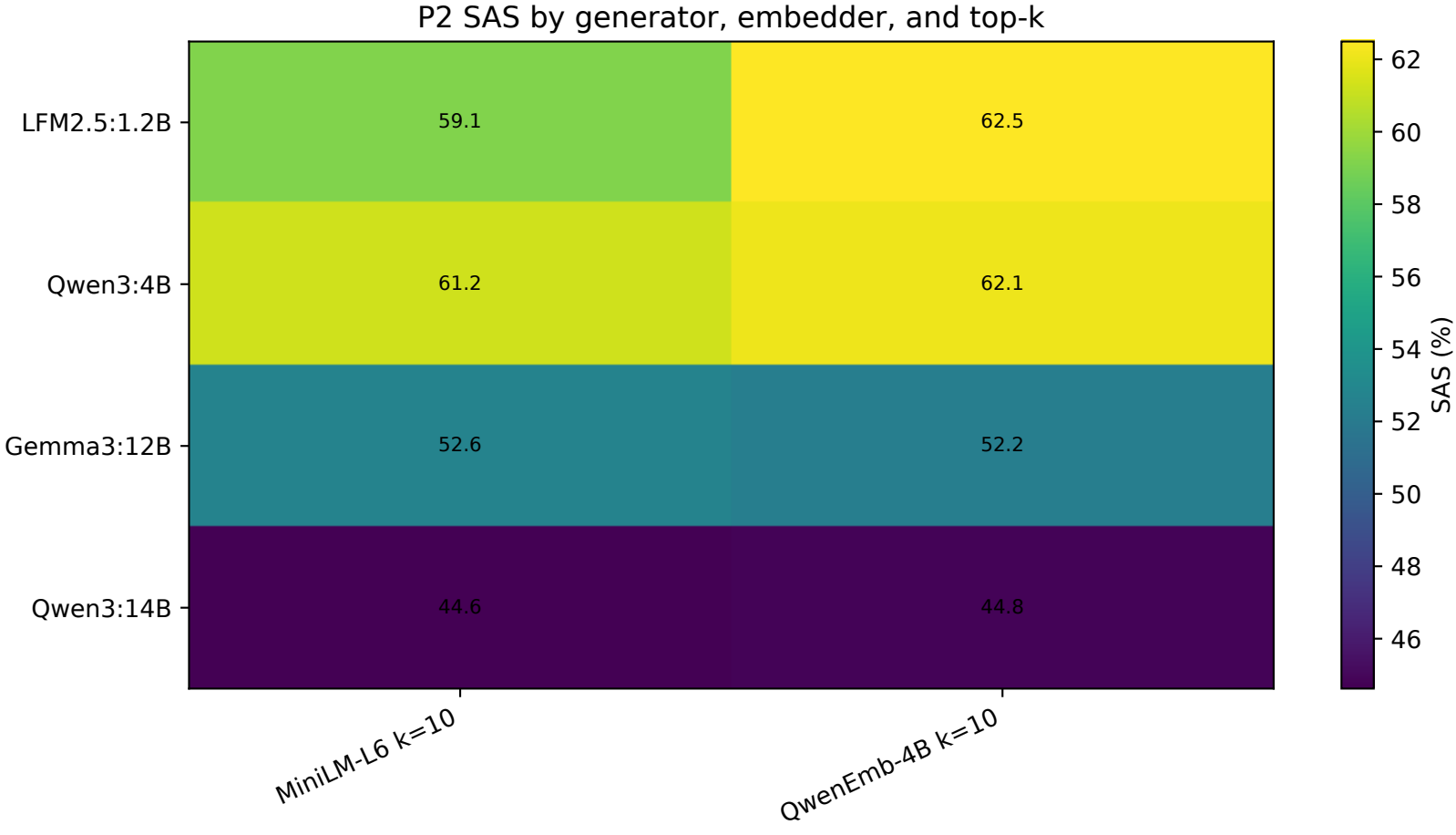

Runtime failure rate by generator and pipeline
P1
P2
P3
Failed trials (%)
0.0
0.2
0.4
0.6
0.8
0.2
0.2
0.1
0.8
0.2
0.9
LFM2.5:1.2B
Qwen3:4B
Gemma3:12B
Qwen3:14B

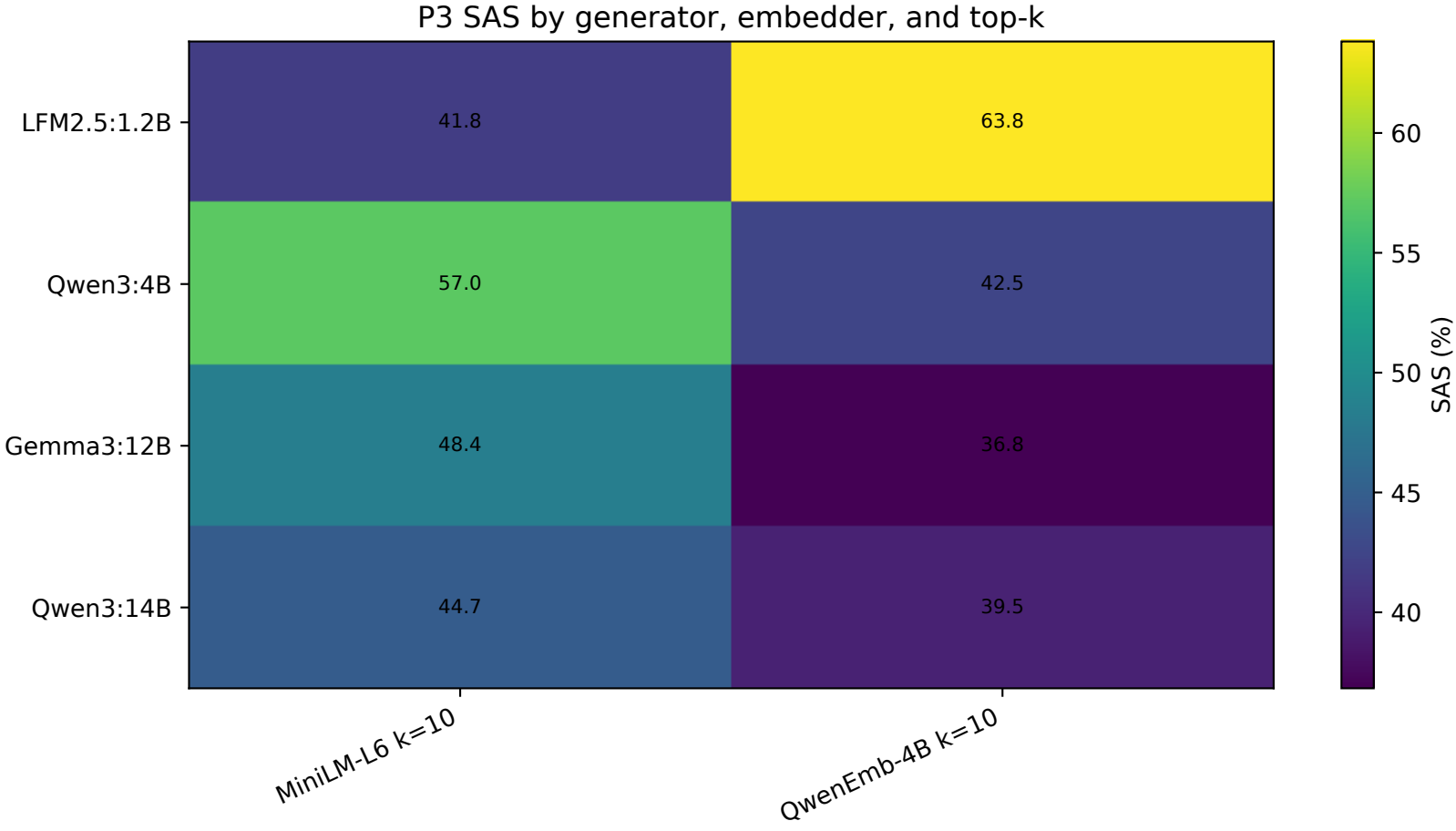

Runtime failure count by generator and pipeline
P1
P2
P3
Failed trials (count)
0
2
4
6
8
1
2
1
4
2
9
LFM2.5:1.2B
Qwen3:4B
Gemma3:12B
Qwen3:14B


Quality-latency trade-off
Qwen3:4B
Qwen3:4B
Gemma3:12B
Gemma3:12B
Gemma3:12B
Qwen3:14B
Qwen3:14B
Qwen3:4B
Qwen3:14B
LFM2.5:1.2B
LFM2.5:1.2B
LFM2.5:1.2B
P1
P2
P3
Best configuration F1 (%)
Mean total time per question (s)
10
20
30
40
50
5
10
15
20
25
30
35


Generator scaling: semantic answer similarity (SAS)
SAS (%)
0
10
20
30
40
50
60
P1
Best P2
Best P3
28.7
62.5
63.8
37.1
62.1
57.0
42.9
52.6
48.4
38.5
44.8
44.7
LFM2.5:1.2B
Qwen3:4B
Gemma3:12B
Qwen3:14B


Semantic vs lexical answer quality
Mean SAS (%)
Mean F1 (%)
60
55
50
45
40
35
30
10
20
30
40
50
LFM2.5:1.2B
Qwen3:4B
LFM2.5:1.2B
Gemma3:12B
Qwen3:4B
Qwen3:14B
Gemma3:12B
Gemma3:12B
Qwen3:14B
Qwen3:14B
Qwen3:4B
LFM2.5:1.2B
P1
P2
P3


P2 SAS by generator, embedder, and top-k
LFM2.5:1.2B
Qwen3:4B
Gemma3:12B
Qwen3:14B
59.1
62.5
61.2
62.1
52.6
52.2
44.6
44.8
MiniLM-L6 k=10
QwenEmb-4B k=10
SAS (%)
62
60
58
56
54
52
50
48
46


P3 SAS by generator, embedder, and top-k
LFM2.5:1.2B
Qwen3:4B
Gemma3:12B
Qwen3:14B
41.8
63.8
57.0
42.5
48.4
36.8
44.7
39.5
MiniLM-L6 k=10
QwenEmb-4B k=10
SAS (%)
60
55
50
45
40

| Generator | P2 gEM-all k = 10 | P3 gEM-all k = 10 |
|---|---|---|
| LFM2.5:1.2B | 0.054 | 0.032 |
| Qwen3:4B | 0.268 | 0.220 |
| Gemma3:12B | 0.156 | 0.130 |
| Qwen3:14B | 0.030 | 0.026 |

| Generator | P2 gEM-hit k = 10 | P3 gEM-hit k = 10 |
|---|---|---|
| LFM2.5:1.2B | 0.062 | 0.046 |
| Qwen3:4B | 0.392 | 0.358 |
| Gemma3:12B | 0.248 | 0.216 |
| Qwen3:14B | 0.054 | 0.054 |

| Generator | P2 gF1-all k = 10 | P3 gF1-all k = 10 |
|---|---|---|
| LFM2.5:1.2B | 0.097 | 0.065 |
| Qwen3:4B | 0.374 | 0.299 |
| Gemma3:12B | 0.307 | 0.256 |
| Qwen3:14B | 0.182 | 0.166 |

| Generator | P2 gF1-hit k = 10 | P3 gF1-hit k = 10 |
|---|---|---|
| LFM2.5:1.2B | 0.123 | 0.093 |
| Qwen3:4B | 0.536 | 0.484 |
| Gemma3:12B | 0.458 | 0.414 |
| Qwen3:14B | 0.267 | 0.257 |

| Generator | P2 gSAS-all k = 10 | P3 gSAS-all k = 10 |
|---|---|---|
| LFM2.5:1.2B | 0.4233982751108706 | 0.3785645865499973 |
| Qwen3:4B | 0.43196949347263885 | 0.35224360428780377 |
| Gemma3:12B | 0.3585431028790772 | 0.2940862665884197 |
| Qwen3:14B | 0.3095918326321989 | 0.28972758180386715 |

| Generator | P2 gSAS-hit k = 10 | P3 gSAS-hit k = 10 |
|---|---|---|
| LFM2.5:1.2B | 0.6147370760403573 | 0.613684478001669 |
| Qwen3:4B | 0.6198277789054211 | 0.5692555258640336 |
| Gemma3:12B | 0.5253669760469347 | 0.47846076981164515 |
| Qwen3:14B | 0.4456327605850426 | 0.4460650558673566 |

| Generator | P1 SAS | P2 SAS k = 10 | P3 SAS k = 10 |
|---|---|---|---|
| LFM2.5:1.2B | 0.2872388741932809 | 0.6249213159624487 | 0.6381698218043893 |
| Qwen3:4B | 0.3705349908115152 | 0.6205507195302713 | 0.5702515269997334 |
| Gemma3:12B | 0.42923363619856536 | 0.5259193611647933 | 0.4839951148852706 |
| Qwen3:14B | 0.384995903301563 47 | 0.44790205637551844 | 0.4466873566871659 |

| Embedder | Top-k | Recall@k (%) | R-Prec (%) | Mean retrieval time (s |
|---|---|---|---|---|
| MiniLM-L6 | 10 | 95.6 | 68.3 | 0.030 |
| QwenEmb-4B | 10 | 95.6 | 68.3 | 2.296 |